

Topic 1: Introduction to Cybersecurity

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
Cybersecurity	The practice of protecting systems, networks, and programs from digital attacks.	Defending computers and data from hackers.	Computer, network mattu data na hackers inda kapadoddu.
CIA Triad	A model that defines the three core principles of cybersecurity: Confidentiality, Integrity, Availability.	Confidentiality (secret aagirbeku), Integrity (correct aagirbeku), Availability (sigbeku).	Secret irbeku, correct irbeku, easy agi access aagirbeku anta 3 main goals.
Information Security	Protecting the confidentiality, integrity, and availability of information, regardless of the form.	Data secure aagirbeku — paper alli irli, computer alli irli.	Yavde data irli, adu secure aagirbeku anta concept.
Cyber Attack	A deliberate exploitation of computer systems, networks, or technologies.	Hacker intentionally attack maadodu system mele.	Namm system mele purposefully attack maadodu.
Asset	Anything of value to an organization (data, hardware, software).	Company ge value iruva item — data, systems, etc.	Important stuff company ge — like server, data, app.
Vulnerability	A weakness in a system that can be exploited by threats.	System nalli iruva loose point.	Hole ide system nalli — adarinda hacker barbahudu.
Threat	Any circumstance or event with the potential to harm information or systems.	Something that may damage or misuse data.	Damage madoke chance ide anta situation.
Risk	The potential for loss or damage when a threat exploits a vulnerability.	$\text{Risk} = \text{Threat} + \text{Vulnerability} + \text{Impact}$.	Hacker barbahudu, hole ide, adinda damage aagbahudu — adu risk.

Topic 2: Cyber Threats and Attacks

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
Threat Actor	An entity responsible for an event that can negatively impact an organization's assets.	Person or group that can attack systems.	Yaaru attack maadthare anta hacker/gang.

Attack Surface	The total number of points where an unauthorized user can try to enter or extract data.	All the ways hacker can get into your system.	Hacker hogbeku andre ella doors ideyo, adu attack surface.
Passive Attack	An attack where the attacker only observes or steals data without altering it.	Just spying or sniffing data, no changes.	Data na just nododu or copy maadodu.
Active Attack	An attack where the attacker modifies or damages systems or data.	Hacker change maadthane or destroy maadthane.	System na break maadodhu, change maadodhu.
Reconnaissance	The process of gathering information about a target system before attacking.	Hacker collect maadthane info about system.	First nodthare – yenu info sigutte anta gather maadthare.
Phishing	Fraudulent attempt to obtain sensitive information by pretending to be a trusted entity.	Fake email/text kottu password hakskolodu.	Fake mail kottu “link click maadi” anta helthare – alli password kodbiditira.
Spear Phishing	Targeted phishing aimed at a specific individual or organization.	Phishing, but only for you or your company.	Ninna hesarige fake mail barutte – nimge thiliyoke try maadthare.
Whaling	A phishing attack targeting high-profile individuals like CEOs.	Special phishing aimed at top bosses.	CEO/manager ge phishing maadodhu.
Social Engineering	Manipulating people into giving confidential info or access.	Tricking humans, not systems.	Manushyana maaye maadodu — info kodsoke.
Brute Force Attack	An attack where every possible password is tried until the right one is found.	Guessing password baruvaregu try maadodhu.	One by one password try maadthare, sigodtanaka.
Dictionary Attack	Using a list of common passwords to break into systems.	Try common passwords first.	Normal aagiru passwords list try maadthare.
Man-in-the-Middle (MITM)	An attacker secretly intercepts communication between two parties.	Hacker sits in between your chat/email.	Ninna message nalli naduve hacker irthare.
Denial of Service (DoS)	Overwhelming a system with traffic to make it unavailable.	Server busy maadthare – so others can’t use.	Site ge jaasti request kodthare – system crash agutte.

Distributed DoS (DDoS)	A DoS attack using many machines to flood a target.	Same as DoS but from many computers.	Network ge chikkakalla computers inda flood maadthare.
SQL Injection	Inserting malicious SQL into a query to control the database.	Hacker injects bad code into app's database.	Form nalli hacker code haakthare – database open aagutte.
Cross-Site Scripting (XSS)	Injecting malicious scripts into websites to run in user browsers.	Site nalli hacker code haakthare, user ge display aagutte.	Website nalli javascript haaki, user nodthaane hacker script.
Zero-Day Attack	An attack exploiting a vulnerability that the vendor doesn't know about yet.	Bug ide, but developer gothilla – hacker use maadthare.	Bug developer ge gothilla, hacker already use maadtane.
Privilege Escalation	Gaining higher-level access than authorized.	Normal user ge admin rights sigodu.	Low level user, admin level ge hogthare.
Supply Chain Attack	Attacking a third-party vendor to reach the real target.	Hacker uses vendor system to get into your system.	Vendor ge attack maadi, neevu ge reach aagthare.

Topic 3: Malware Types

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
Malware	Malicious software designed to harm, exploit, or disable computers or systems.	Any bad software made to damage or steal data.	System ge haanikara software – hacker haakthare.
Virus	Malware that attaches itself to a file/program and spreads when executed.	Spreads only when the infected file is opened.	Infected file open maadidre matra spread aagutte.
Worm	A self-replicating program that spreads without human action.	Spreads on its own, no need to open.	Thane thane spread aagutte – open maadbekalla.
Trojan Horse	Malware disguised as a legitimate program.	Fake good-looking software that's actually harmful.	"Useful app" tara irutte, but hacker software irutte.
Ransomware	Malware that encrypts files and demands payment to unlock them.	Locks your data and asks money to unlock.	Ninna files lock maadi, haana kodre open maadtare.
Spyware	Software that secretly collects user data.	Watches what you're doing and sends to hacker.	Ninna activity silent aagi nodthare, data hacker ge kodthare.

Adware	Automatically delivers unwanted ads to users.	Annoying ads that pop up often.	Baekagilla ads barutte, disturb maadutte.
Rootkit	Software tools used to hide malicious activities or give unauthorized root access.	Helps hackers hide inside your system.	System nalli hide aagi, admin access kodthare hacker ge.
Keylogger	Software that records every key pressed on a keyboard.	Captures what you type, including passwords.	Keyboard ge yen type maadtira, adella record aagutte.
Botnet	A group of infected computers controlled by a hacker.	Hacker controls many systems remotely.	Jaasti infected PCs combine aagi, hacker ge obey maadtave.
Fileless Malware	Malware that doesn't use files, hides in memory, and avoids detection.	Doesn't leave traces on disk – hard to detect.	System nalli trace illa – memory nalli hide aagutte.
Logic Bomb	Malicious code that triggers when certain conditions are met.	Waits for specific time or event to activate.	Time/condition baruvaregu silent – then burst aagutte.
Backdoor	A hidden entry point that bypasses normal authentication.	Secret way to get into a system.	Secret door irutte hacker ge, admin baralla anta.
Scareware	Tricks users into downloading malicious software by scaring them.	Fake alerts like “virus detected” to make you install.	“Danger! Virus ide!” anta scare maadi, hacker app install maadthare.

Topic 4: Network Security

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
Firewall	A network security device or software that monitors and filters incoming/outgoing traffic based on rules.	Controls what data comes in/out of your system.	Yaav info barbahudu, hogbahudu anta gatekeep maadodu.
VPN (Virtual Private Network)	A secure, encrypted connection over a public network to protect data transmission.	Makes your internet connection private and secure.	Public network use maadta iddru, connection secret aagutte.
IDS (Intrusion Detection System)	A system that monitors network traffic for suspicious activity and issues alerts.	Just detects and reports unusual activity.	Yaavdru hacking try aagta ide anta helthade.

IPS (Intrusion Prevention System)	Similar to IDS but also actively blocks threats.	Detects and blocks malicious activity.	Hacker barlilla anta door close maadhade.
Proxy Server	An intermediary server that separates end users from the websites they browse.	Acts like a middleman for browsing safely.	Neenu site nodta idiya? Adakke proxy na use maadhare.
DMZ (Demilitarized Zone)	A separate network zone that adds an extra layer of security between public and internal networks.	Buffer zone between public internet and private network.	Public network matthu private network naduve neutral zone.
Network Segmentation	Dividing a network into segments to improve security and performance.	Cuts network into pieces so hacker can't go everywhere.	Hacker baradre full network access sigalla, block maadbahudu.
VLAN (Virtual LAN)	A logical subdivision of a physical network to group devices separately.	Like dividing office network into teams.	Network na teams tara divide maadoddu.
NAT (Network Address Translation)	Maps private IPs to a public IP, hiding internal IPs from the internet.	Helps hide your real internal IP from outsiders.	Real IP hide maadhade, public IP ge map maadhade.
Port Scanning	A technique used to identify open ports and services on a target system.	Checking which doors are open in a system.	Yaav doors open ide anta nodthare hacker.
Packet Sniffing	Capturing data packets moving through a network.	Spying on data moving in the network.	Network nalli hogta iruva data na spy maadoddu.
MAC Filtering	Allowing or denying devices based on their MAC address.	Network ge yaav device connect aagbahudu anta MAC ID mele decide maadoddu.	Phone/Laptop MAC ID check maadi allow maadoddu.
TLS/SSL	Cryptographic protocols for secure communication over a network.	Used for HTTPS (secure websites).	Website secure agbeku andre TLS/SSL beku.
Port Security	Controlling access to a network by limiting the devices that can connect to switch ports.	Prevents unauthorized devices from plugging in.	Switch ge yaav device connect aagbeku anta control maadoddu.
Wi-Fi Encryption (WPA2/WPA3)	Security standards for wireless networks.	Keeps Wi-Fi connection safe from hackers.	Wi-Fi ge password hakidare secure aagbeku – WPA2/WPA3 beku.

Topic 5: Identity and Access Management (IAM)

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
IAM (Identity and Access Management)	A framework of policies and technologies to ensure the right individuals access the right resources at the right time.	Controls who can access what, when, and how.	Yaav ogaru yaav file open maadbahudu anta control maadodu.
Authentication	Verifying the identity of a user or system.	Checking who you are.	Nivu yaaru anta check maadodu – login process.
Authorization	Granting access rights based on identity and policies.	What you're allowed to do after login.	Login aadmele yaav folder access sigutte anta nododdu.
MFA (Multi-Factor Authentication)	Requires two or more verification factors to gain access.	Not just password – OTP, fingerprint, etc.	Password + OTP + Fingerprint idre jaasti secure.
SSO (Single Sign-On)	Allows users to access multiple systems using one login.	One login → access to many apps.	Ondu sala login aadre, ella app access sigutte.
RBAC (Role-Based Access Control)	Access rights are assigned based on user roles.	Access depends on job role (admin, user, HR, etc).	Admin ge full access, intern ge limited anta divide maadoddu.
ABAC (Attribute-Based Access Control)	Access is based on attributes (like department, location, time).	More flexible than RBAC, checks many conditions.	Location/time/role ella consider maadi access kododdu.
LDAP (Lightweight Directory Access Protocol)	A protocol used to access and maintain distributed directory info services.	Used in organizations for user directory and login info.	Company users' details store maadoke use maadthare.
Federated Identity	Linking a single identity across multiple systems (cross-org access).	One ID works across different companies' systems.	One login use maadi, beda systems nalli access barutte.
Biometric Authentication	Authentication using physical traits like fingerprint, retina, or voice.	Login using your body features.	Thumb scan, face ID, voice ge base aad login.
Access Control List (ACL)	A list that defines permissions for users or systems on resources.	Tells who can read/write/execute a file.	Yaar access maadbodu anta list iddu nodthade.
Identity Federation	Trust established between different domains to share authentication.	Example: Google login for third-party sites.	Google ID use maadi beda site login aagoke.

Privileged Access Management (PAM)	Special controls for users with high-level system access.	Admin users ge extra security.	High access iruva janru ge jaasti secure policy beku.
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Topic 6: Cryptography

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
Cryptography	The science of securing data through encoding techniques to ensure confidentiality, integrity, and authenticity.	Art of protecting data by converting it to unreadable form.	Data na bejaar agi change maadi, hacker ge artha agalla anta secure maadoddu.
Encryption	The process of converting plaintext into ciphertext to prevent unauthorized access.	Turns readable data into unreadable.	Normal data na secret code aagi maarthare.
Decryption	Converting encrypted data back into its original form.	Making secret data readable again.	Code na thirugi artha baruva haage maadoddu.
Plaintext	Original readable data before encryption.	Normal text/data.	Encrypt madlilla data – as it is.
Ciphertext	Encrypted, unreadable form of data.	Encrypted version of data.	Data secret code haage irutte.
Key	A piece of information used in cryptographic algorithms to encrypt or decrypt data.	Secret value used to lock/unlock data.	Data lock maadoke key, open maadoke same key beku.
Symmetric Encryption	Uses the same key for both encryption and decryption.	Same key to lock and unlock.	Ondey key na use maadthare encode/matthu decode ge.
Asymmetric Encryption	Uses a pair of keys: public key for encryption and private key for decryption.	Public key locks, only private key unlocks.	Ondu key share maadbahudu (public), bere key hide maadbeku (private).
Public Key	Shared key used to encrypt data.	Anyone can use to send you secret data.	Public ge kodbahudada key – but open maadoke private beku.
Private Key	Secret key used to decrypt data.	Only you have it, used to open messages.	Secret key – ninna message open maadoke use maadthira.
Hashing	A one-way process that converts data into a fixed-size hash value.	Can't reverse it; used for passwords, checksums.	Data na short code aagi maadthare – back maadakke agalla.

Salt	Random value added to passwords before hashing to increase security.	Extra security to prevent password cracking.	Password ge masale haakidre haage – different aagutte code.
Digital Signature	A cryptographic value that verifies the authenticity and integrity of data.	Confirms data is from right sender, not modified.	Message original idiya anta signature haage check maadbahudu.
Certificate	A digital document used to prove ownership of a public key.	Issued by Certificate Authority (CA).	HTTPS site ge namma trusted key ide anta certificate.
PKI (Public Key Infrastructure)	Framework for managing digital keys and certificates.	System that handles public/private keys and trust.	Key/certificate create, store, validate maadoke system.
SSL/TLS	Protocols for securing communications over a network using encryption.	Used for HTTPS websites.	Site secure agbeku andre SSL/TLS irbeku.
Nonce	A number used once in cryptographic communication to prevent replay attacks.	Prevents reuse of old messages.	Onde sala use maadoke number – old messages baralla.

Topic 7: Security Governance, Risk, and Compliance (GRC)

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
GRC (Governance, Risk & Compliance)	A structured approach to align IT with business goals, manage risk, and meet regulatory requirements.	Ensures organization follows rules, manages risks, and works securely.	Company proper rules follow maadta idiya, risk handle maadta idiya anta check maadodu.
Governance	The framework of rules, practices, and processes by which a company is directed and controlled.	Making security part of business decisions and strategy.	Business nalli security yenu rules idiya anta nododdu.
Risk Management	The process of identifying, assessing, and controlling threats to an organization's assets.	Spotting and handling possible future security problems.	Future nalli yaav risk barbahudu anta nodi plan haakoddu.
Risk Assessment	Identifying and evaluating potential risks to assets and operations.	What could go wrong and how bad would it be?	Yenu thappu aagbahudu anta calculate maadoddu.
Risk Mitigation	Steps taken to reduce the impact of risks.	Reducing how much a risk can hurt.	Risk irbahudu, aadre adara haanikke kadime maadoddu.

Compliance	Following laws, regulations, standards, and internal policies.	Making sure you're doing everything legally and correctly.	Government/industry rules ella follow maadbeku.
Policy	A formal document that outlines an organization's principles, rules, and expectations for security.	Written rules about how to handle security.	Company nalli yenu maadbekaagide anta rule book.
Standard	Specific requirements or specifications for systems and processes.	Step-by-step guidelines to meet policies.	Policy antha general, standard antha procedure.
Procedure	Detailed instructions to achieve a specific task or requirement.	Exact how-to for each security activity.	Ondu work na yenge maadbekaagide anta step-by-step.
Control	A safeguard or countermeasure to reduce risk.	Actions taken to prevent or fix security issues.	Risk avoid maadoke use aaguvudu.
Audit	A formal examination of how well policies and controls are being followed.	Official checking if rules are followed.	Company nalli yaaru rules follow maadidare anta check maadodu.
Security Framework	A set of best practices and guidelines for implementing security (e.g., NIST, ISO 27001).	Reference for doing security the right way.	Model system security ge – steps haage plan irutte.
ISO 27001	International standard for Information Security Management System (ISMS).	Global guideline for how to manage and protect data.	Global level security certificate haage – ISO irbeku.
NIST	U.S. government framework for cybersecurity standards and best practices.	Strong security guideline widely followed.	US government standard – many company use maadthare.
Data Classification	Categorizing data based on sensitivity and importance.	Deciding which data is secret and needs more protection.	Yaav data jaasti secret anta tag haakoddu.
Acceptable Use Policy (AUP)	Rules on how employees can use company tech resources.	What you can/can't do with office systems.	Company computer yenu use maadbekaagide anta rules.
Incident Response Plan (IRP)	A documented process to detect, respond to, and recover from security incidents.	What to do if a security problem happens.	Hacker attack aadmele yenu maadbekaagide anta ready plan.
Business Continuity Plan (BCP)	Strategy to keep business running during major disruptions.	Keep work going even if there's a disaster.	Office nalli issue aadru, work stop aagalla haage backup plan.

Disaster Recovery Plan (DRP)	Steps to restore systems after a disaster.	How to get systems back after crash.	Server down aadmele, yenge repair maadbeku anta plan.
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Topic 8: Security Testing & DevSecOps

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
DevSecOps	Integration of security practices into DevOps processes throughout the software development lifecycle.	Security is included from start to finish in app development.	App build madoke startinda security include maadoddu.
SDLC (Software Development Life Cycle)	A process for planning, creating, testing, and deploying software.	Step-by-step flow of how software is developed.	Software yenge build aagutte anta full cycle.
SAST (Static Application Security Testing)	Examines source code or binaries for vulnerabilities without executing the program.	Checks code for issues without running it.	Code run maadadele bug/barbodu anta check maadoddu.
DAST (Dynamic Application Security Testing)	Tests the running application to find vulnerabilities during execution.	Runs the app and tries to find security flaws.	App open maadi hacker haage test maadoddu.
IAST (Interactive Application Security Testing)	Combines SAST and DAST using runtime agents during testing.	Live app nalli code + behavior na test maadoddu.	App running time nalli, code + output check maadthade.
RASP (Runtime Application Self-Protection)	Security tech that runs inside an app and detects/block attacks in real time.	App ge brain haaki, itself protect maadthade.	App nalli thaaney protect maadoke logic irutte.
Code Review	Manual or automated inspection of source code to identify bugs or security issues.	Developers checking each other's code.	Code baradmele inond programmer check maadthare.
Secure Coding	Writing code using best practices to prevent vulnerabilities.	Writing safe code from the beginning.	Bug barolla haage clean code baribeku.
CI/CD (Continuous Integration / Continuous Deployment)	DevOps practice of automating code building, testing, and deploying.	Code auto test and release aagutte.	Developer code push maadidre, app auto update aagutte.
Security in CI/CD	Embedding security checks into CI/CD pipelines.	Auto scan + security testing before deploy.	App deploy aagoke munche, hacker check automatic aagutte.

Dependency Scanning	Checking external libraries for vulnerabilities.	Finds issues in 3rd-party packages.	Bera libraries nalli bug ideya anta check maadoddu.
Container Security	Protecting containerized environments like Docker and Kubernetes.	Securing apps running in containers.	Docker/K8s apps secure aagirbeku anta nododdu.
Secrets Management	Secure handling of passwords, API keys, tokens, etc.	Don't hardcode passwords; store them safely.	Code nalli password baribeda, secure store maadbeku.
Fuzz Testing (Fuzzing)	Inputting random data into an application to find crashes or bugs.	Throw junk data to see what breaks.	App ge nonsense data haaki nodthare enu crash aagutte anta.
Penetration Testing (Pentest)	Simulated attack on a system to find exploitable vulnerabilities.	Ethical hacking to test defenses.	Hacker haage attack try maadi loophole sigutta anta nododdu.
Bug Bounty	A program offering rewards to ethical hackers for finding bugs.	Hacker ge paisa kodthare if they report vulnerabilities.	Hacker na kelthare – yenu bug sigidre kodtheeni anta.

Topic 9: Security Tools & Monitoring

Term	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
SIEM (Security Information and Event Management)	Software that collects, analyzes, and correlates logs from various systems in real time for threat detection.	Centralized system to monitor and alert on suspicious activities.	Ella log collect maadi, yenu suspicious aagide anta heli alert maadoddu.
SOAR (Security Orchestration, Automation, and Response)	A platform that automates incident response and integrates multiple security tools.	Auto fix or escalate security issues.	Alert banda mele auto action tegedukoLuva system.
SOC (Security Operations Center)	A dedicated team or place to monitor, detect, analyze, and respond to cybersecurity incidents.	Security control room.	Cyber police station haage – team full time alert irutte.
EDR (Endpoint Detection and Response)	Monitors endpoints (like laptops, desktops) for suspicious behavior and provides response tools.	Detects and responds to threats on user devices.	Laptop/Desktop nalli enadru doubttaadre catch maadoddu.
XDR (Extended Detection and Response)	Unified security incident detection and response across multiple layers (endpoint, network, cloud).	All-in-one monitoring system.	Endpoint + network + cloud ella nododdu.

NDR (Network Detection and Response)	Monitors network traffic for suspicious patterns and threats.	Looks for unusual behavior in network.	Network nalli yaar enu strange maadta iddare anta detect maadoddu.
Threat Intelligence	Knowledge about existing and emerging threats to help detect and respond to attacks.	Info about hacker strategies and malware.	Hacker yenu plan maadta iddane anta news haage.
Threat Hunting	Proactively searching for threats in systems before alerts are triggered.	Finding hidden threats before damage.	Hacker sigoke munche hudukoddu – clue haage.
Log Management	Collecting, storing, and analyzing system logs for security monitoring.	Keeping track of all activity.	Yaar yenu maadidru anta record ittu check maadoddu.
UEBA (User and Entity Behavior Analytics)	Uses machine learning to detect unusual behavior of users and systems.	Finds suspicious behavior automatically.	Namma normal usage change aadre alert maadthade.
Anomaly Detection	Identifying behavior that deviates from normal patterns.	Spotting weird stuff happening.	Usual alla andre – system doubt maadthade.
Incident Management	Process of handling and resolving security incidents.	What to do when something goes wrong.	Attack aadmele yenu action thegiyodu anta ready irbeku.
Playbook (Security)	Pre-defined steps for handling specific types of security incidents.	Checklist for incident response.	Yenu aadrallu nodi, step by step solve maadoke guide.
Alert Fatigue	Too many alerts causing analysts to miss real threats.	Too many false alarms leading to ignoring real ones.	Jaasti alerts bandre – true alert ignore aagbahudu.
SIEM Correlation Rules	Set of conditions that combine multiple logs/events to detect threats.	If A + B + C happen, alert as attack.	Ondond alert mix maadi hacker detect maadoke rule.

Types of Authentication

✅ **Authentication = "Who are you?"**

It's the process of **verifying the identity** of a user or system.

Authentication Type	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
Password-based Authentication	Verifies identity using a secret word or phrase (password).	Most common method – you enter your password to log in.	Neevu password haaki login maadodu.
PIN-based Authentication	Uses a personal identification number to authenticate the user.	Similar to password but shorter (like ATM PIN).	ATM PIN haakidre haage – short code use maadoddu.

Biometric Authentication	Uses unique biological traits like fingerprint, face, or retina.	Your body becomes the password.	Ninna finger/thumb/face na use maadi login maadoddu.
Token-based Authentication	Involves a physical or digital token (e.g., OTP, smartcard).	You get a code/token that proves who you are.	Phone ge OTP bandre haage – adu token aagutte.
Multi-Factor Authentication (MFA)	Combines two or more types of authentication (e.g., password + OTP).	Extra security by mixing multiple methods.	Password + OTP haaki login maadoddu – double layer.
Two-Factor Authentication (2FA)	A subset of MFA using exactly two different methods.	Mostly password + phone OTP combo.	Ondu password, innondu OTP – rendu steps irutte.
Certificate-based Authentication	Uses digital certificates (like X.509) to verify identity.	Often used in enterprises and servers.	System ge certificate kodsi – adu identity proof.
Single Sign-On (SSO)	One login gives access to multiple applications.	Login once, use many apps (like Google SSO).	Ondey sala login maadidre, ella apps open aagutte.
Federated Identity	Shares identity between multiple systems across organizations.	Login on one system works on another trusted one.	One website nalli login madidre, bere site kooda accept maadutte.
OAuth (Open Authorization)	Allows one site to use your identity from another (e.g., login with Google).	Secure way to share login across apps.	Login with Google/Facebook anta nodiddeya? adu OAuth.
Behavioral Authentication	Verifies identity based on typing pattern, mouse usage, etc.	AI checks your usage style.	Yenge type maadtira, adara pattern bagge AI nodthade.
Risk-based Authentication (RBA)	Dynamically adjusts authentication level based on user's risk profile.	If login looks suspicious, asks for more verification.	Dubaayinda login madidre extra OTP kelthade.
Passwordless Authentication	Uses biometrics, magic links, or push notifications instead of passwords.	No password needed at all.	Password beda – face, fingerprint or phone nalli approve maadoddu.

Types of Authorization

✅ **Authorization = "What are you allowed to do?"**

After authentication, authorization **controls access** to resources or actions.

Authorization Type	Technical Definition	General Explanation	Simplified Kannada (Transliteration)
Role-Based Access Control (RBAC)	Access is granted based on user roles (e.g., admin, user).	Roles decide what you can access.	Ninna job role (admin/user) prakara access sigutte.

Attribute-Based Access Control (ABAC)	Access based on attributes of user, environment, or resource.	More dynamic than roles (e.g., time, location, dept).	Role alla, yaaru, elli iddare anta attributes base maadi access kodthare.
Discretionary Access Control (DAC)	Resource owners decide who gets access.	You control who can access your files.	Ninna file ge yaar access irbeku anta neevu decide maadtira.
Mandatory Access Control (MAC)	Access is enforced by system policies and not user-controlled.	High-security environments like military.	System decide maadutte – user control illa.
Policy-Based Access Control (PBAC)	Access is managed using predefined policies written in code or rules.	Used in cloud and enterprise systems.	System nalli rule haakidre, adara base mele access sigutte.
Rule-Based Access Control	Access based on specific system rules.	Similar to PBAC but rule-focused.	Simple rules haaki – if this, then allow/block.
Time-Based Authorization	Grants access only during specific time windows.	Used for shift-based users or temporary access.	Office hours ge access irutte, night ge block maadthare.
Location-Based Authorization	Access control based on user's physical or IP location.	Only allow if user is in India, for example.	India nalli idre allow, bere desha inda block maadoddu.
Context-Based Authorization	Combines multiple conditions like time, device, and location.	Smarter access decisions.	Time + device + place kooda consider maadthare.
Fine-Grained Authorization	Allows precise control over specific data and actions.	Like row-level access in a database.	Very detailed – one table row kooda control maadbahudu.
Delegated Authorization	User A can temporarily give access to User B.	Give someone else your access for a while.	Ninna permission inobbarige kodbahudu – temporary aagi.

Cybersecurity –Important Terms

#	Term	Technical Definition	Example
1	Cybersecurity	Protection of internet-connected systems from cyber threats.	Securing apps, data, and networks.
2	Firewall	Controls traffic between trusted and untrusted networks.	Home router blocks suspicious traffic.
3	Antivirus	Detects and removes viruses from systems.	Windows Defender removing malware.
4	Malware	Malicious software designed to damage or gain access.	Virus, Worms, Trojans.

5	Ransomware	Locks data and demands ransom for release.	WannaCry attack.
6	Phishing	Deceptive method to steal sensitive data via fake identity.	Fake bank login emails.
7	Spear Phishing	Targeted phishing attack on a specific person.	CEO gets fake invoice email.
8	Encryption	Securing data by converting it into unreadable format.	WhatsApp messages.
9	Decryption	Reversing encrypted data back to readable form.	Password decrypt.
10	Authentication	Proving identity of a user.	Login with username and password.
11	Authorization	Giving rights to access resources.	Only admin can delete users.
12	Two-Factor Authentication (2FA)	Requires two forms of identity proof.	Password + OTP.
13	Multi-Factor Authentication (MFA)	More than two verification methods.	Password + OTP + fingerprint.
14	Access Control	Mechanism to allow/deny user access.	Role-based file access.
15	Zero Trust	Never trust, always verify access model.	All access checked regardless of location.
16	Patch Management	Regularly updating software to fix bugs.	Microsoft Patch Tuesday.
17	Vulnerability	Weakness that can be exploited.	Unpatched software bug.
18	Exploit	Code or tool used to take advantage of vulnerability.	Remote code execution.
19	Threat	Potential danger to data or systems.	Malware, phishing.
20	Risk	Probability of a threat exploiting a vulnerability.	Weak passwords can be guessed.
21	Mitigation	Reducing risk through controls.	Enable firewall.
22	Red Team	Offensive security experts who simulate attacks.	Ethical attackers test defenses.
23	Blue Team	Defensive team that protects infrastructure.	Detect and stop threats.
24	Reconnaissance	Gathering information about the target.	Scanning IPs.
25	Beaconing	Periodic contact from compromised system to attacker.	Malware reporting back to server.
26	SIEM	Security Information and Event Management system.	IBM QRadar, Splunk.

27	IDS (Intrusion Detection System)	Monitors network for malicious activity.	Alerts on strange traffic.
28	IPS (Intrusion Prevention System)	Blocks threats in real time.	Stops DDoS traffic.
29	DDoS	Overloading system with traffic.	Server crash via botnet.
30	Botnet	Network of compromised systems used maliciously.	IoT botnet attack.
31	Man-in-the-Middle (MITM)	Attacker intercepts communication between parties.	Fake Wi-Fi capturing login.
32	Spoofing	Faking identity or source.	Fake email sender.
33	Social Engineering	Tricking people to give up confidential data.	Fake IT support calls.
34	Insider Threat	Threat from within the organization.	Employee leaks data.
35	Data Breach	Unauthorized data exposure.	Customer info leak.
36	Digital Forensics	Recovering and analyzing digital evidence.	Hacked phone analysis.
37	SOC (Security Operations Center)	Team that monitors and defends infrastructure.	24x7 security team.
38	Security Policy	Documented rules for protecting systems.	Password strength rules.
39	Security Audit	Assessment of systems for vulnerabilities.	Pen-test review.
40	Penetration Testing	Ethical hacking to test security.	Simulated attack.
41	Threat Modeling	Identifying possible threats in a system.	STRIDE model.
42	Cyber Hygiene	Basic practices to maintain security.	Changing passwords.
43	Privileged Access	Special high-level access to systems.	Admin accounts.
44	Least Privilege	Giving minimum access required.	Interns can't delete data.
45	Hashing	Converting data to fixed-length code for storage.	Password hash.
46	Tokenization	Replacing sensitive data with unique codes.	Tokenized credit card data.
47	Sandboxing	Isolating suspicious programs.	Virtual box testing.
48	Whitelisting	Allowing only approved apps or users.	Only Outlook can open.
49	Blacklisting	Blocking known bad apps or users.	Spam sites.
50	Incident Response	Process to handle a cyberattack.	Isolate and recover.
51	Security Awareness Training	Educating users about security practices.	Phishing simulations.
52	Certificate Authority (CA)	Issues digital certificates for website security.	HTTPS uses SSL certificate.

53	SSL/TLS (Secure Socket Layer / Transport Layer Security)	Encrypts web data transmission.	HTTPS websites.
54	Endpoint Security	Protecting individual devices like PCs, phones.	Antivirus, EDR tools.
55	Cloud Security	Protecting cloud-based resources.	AWS, Azure policies.
56	Mobile Device Management (MDM)	Controls and secures mobile devices in a network.	Erase data if phone lost.
57	Supply Chain Attack	Attack via third-party software.	SolarWinds attack.
58	Zero-Day Attack	Exploiting a vulnerability before it's fixed.	Unknown software bug used in attack.
59	Credential Stuffing	Using leaked login info on multiple sites.	Netflix account tried on Amazon.
60	Cyber Resilience	Ability to continue operations after an attack.	Business continuity after ransomware.

Ethical Hacking –Important Terms

#	Term	Technical Definition	Example
1	Ethical Hacking	Authorized testing of systems to find and fix vulnerabilities.	Company hires hacker to test their app.
2	Hacker	Person skilled in computer systems who can bypass protections.	Ethical hacker breaks into a test system.
3	White Hat	Ethical hacker who works legally to improve security.	Bug bounty professionals.
4	Black Hat	Hacker who breaks systems for illegal purposes.	Stealing bank data.
5	Gray Hat	Hacks without permission but without malicious intent.	Finds a bug and informs the company.
6	Penetration Testing (Pentest)	Simulated attack to test system security.	Red team tests company's network.
7	Reconnaissance	Information gathering phase before an attack.	Whois, Google dorking.
8	Footprinting	Collecting detailed info about the target.	Finding IPs, DNS, emails.
9	Scanning	Identifying open ports and services on a system.	Nmap used to scan network.
10	Enumeration	Extracting detailed info from scanned systems.	Listing users from a server.

11	Vulnerability Assessment	Identifying and classifying security weaknesses.	Nessus scan.
12	Exploitation	Taking advantage of a vulnerability to gain access.	SQL injection.
13	Payload	Code that runs after exploitation.	Meterpreter shell.
14	Metasploit	Popular framework for penetration testing.	Used to deliver payloads.
15	Reverse Shell	Hacker gets control of victim's system.	Victim system opens connection to attacker.
16	Privilege Escalation	Gaining higher-level access on a system.	From normal user to admin.
17	Remote Code Execution (RCE)	Running attacker's code on a remote system.	Hack server to run malware.
18	Command Injection	Inserting OS commands into inputs.	Typing ; ls into a form field.
19	SQL Injection	Injecting SQL code into input fields.	' OR '1'='1 bypasses login.
20	Cross-Site Scripting (XSS)	Injecting malicious scripts into web pages.	Pop-up box appears on page.
21	Cross-Site Request Forgery (CSRF)	Forcing a user to perform unwanted actions.	Auto-clicking links using cookies.
22	Brute Force Attack	Trying many passwords to gain access.	Guessing Gmail password.
23	Dictionary Attack	Using a list of common passwords to guess login.	password123, admin.
24	Pharming	Redirecting users from real websites to fake ones.	Fake Paytm page.
25	Social Engineering	Manipulating people to reveal secrets.	Pretending to be IT support.
26	Shoulder Surfing	Watching over someone's screen or keyboard.	Seeing PIN at ATM.
27	Keylogger	Tool that records keystrokes.	Captures passwords.
28	Session Hijacking	Taking over an active user session.	Stealing cookies.
29	Sniffing	Monitoring network traffic.	Wireshark capturing packets.
30	ARP Spoofing	Sending fake ARP messages to intercept traffic.	Redirecting data in LAN.
31	DNS Spoofing	Redirecting domain to fake IP address.	www.google.com -> hacker IP.
32	Eavesdropping	Secretly listening to network communication.	Capturing VoIP call.
33	Man-in-the-Middle (MITM)	Intercepting communication between two systems.	Fake Wi-Fi router.
34	Payload Delivery	Process of sending exploit code to a system.	Using phishing email.
35	Trojan Horse	Malware disguised as useful software.	Fake PDF file.

36	Logic Bomb	Malicious code triggered by a condition.	On a specific date.
37	Rootkit	Software to hide malicious activity.	Hides from antivirus.
38	Backdoor	Secret way to access a system.	Developer left hidden login.
39	Zombie	Infected computer used in botnet.	DDoS attack machine.
40	Botnet	Group of compromised systems.	1000 PCs launching attack.
41	Banner Grabbing	Collecting service/version info from a system.	Apache version from port 80.
42	Port Scanning	Identifying open ports on a system.	Nmap scan.
43	OSINT (Open Source Intelligence)	Gathering public data for hacking.	LinkedIn, social media.
44	Bug Bounty	Reward for finding security flaws.	HackerOne programs.
45	Exploit Kit	Toolkits to exploit multiple vulnerabilities.	Blackhole kit.
46	Credential Dumping	Extracting stored usernames/passwords.	Dumping Windows hashes.
47	Pass-the-Hash	Using password hashes to authenticate.	No password needed.
48	Lateral Movement	Moving through a network after compromise.	From one PC to another.
49	Persistence	Maintaining access even after reboot.	Adding registry keys.
50	Covering Tracks	Deleting evidence of attack.	Log file deletion.
51	Cyber Kill Chain	Phases of a cyber attack.	Recon, weaponize, deliver...
52	Payload Obfuscation	Hiding payload to bypass defenses.	Encryption or changing signatures.
53	Wi-Fi Cracking	Breaking into wireless networks.	Cracking WPA2 password.
54	Fake Access Point	Rogue Wi-Fi set by hacker.	Free airport Wi-Fi.
55	Aircrack-ng	Tool to hack Wi-Fi passwords.	Used in Kali Linux.
56	Kali Linux	Hacking-focused operating system.	Used by ethical hackers.
57	Burp Suite	Web vulnerability scanner and manipulator.	Edit login requests.
58	OWASP Top 10	List of most critical web app vulnerabilities.	XSS, SQLi included.
59	Session Fixation	Forcing victim to use a known session ID.	URL-based attack.
60	Ethical Disclosure	Reporting vulnerabilities responsibly.	Informing company before going public.

What is Repudiation in Cybersecurity? Repudiation means a situation where a user performs some action (like deleting, modifying, or accessing data) but later denies doing it and system cannot prove they actually did it. 🙌 It's the act of "denying responsibility" for an action, and it's a security concern if the system lacks proper logs or evidence.